

Q.4a (20 marks): What are genetic markers and what is their usefulness? Why are blood groups considered as good ...

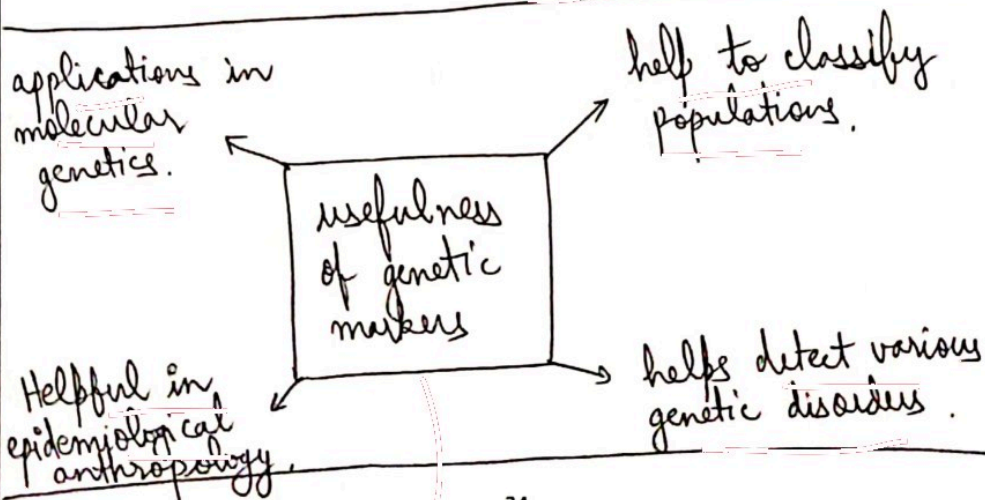
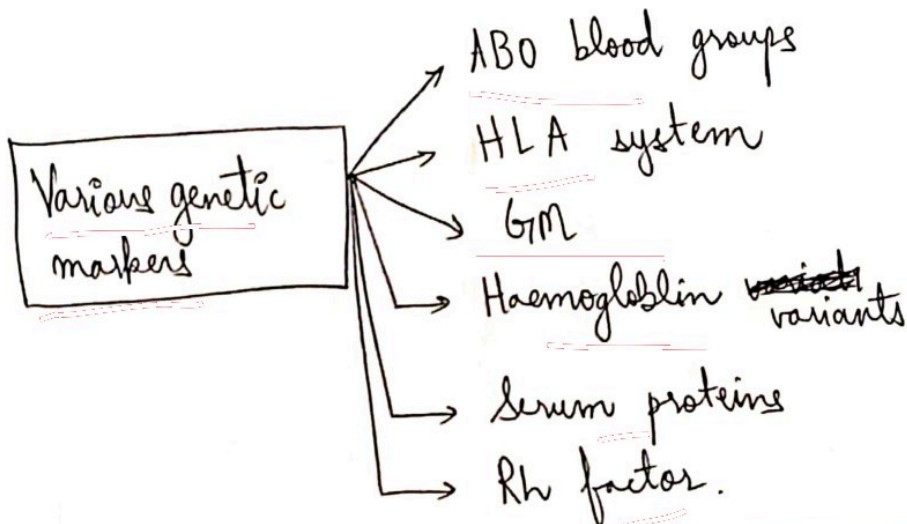
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Q4. (a) What are genetic markers and what is their usefulness? Why are blood groups considered as good genetic markers. Illustrate with examples.

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According to Crawford (1970), genetic markers are genetic entities that are used to classify the population by the presence & absence of their frequency in various organisms.



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Evolutionary studies

Understand migratory patterns

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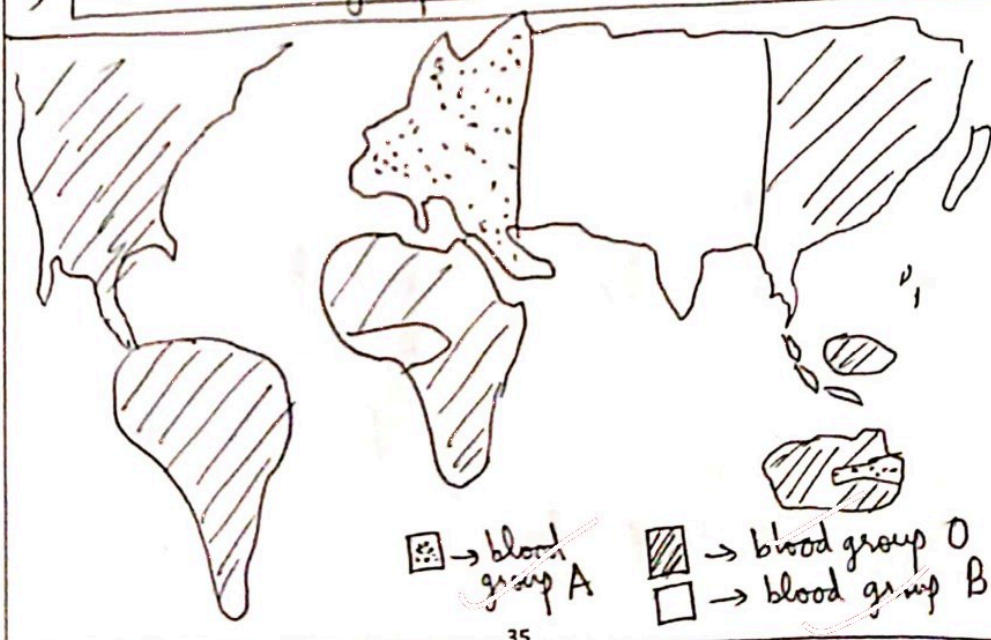
ABO Blood groups are good genetic markers

- 1) Their frequency is very high in world population.
- 2) They are easy to detect.
- 3) They are unaffected by external environment
- 4) They have huge applications in forensic anthropology.

Examples

1) ABO blood group

Mention studies of Weiner (1946-48) on gene frequencies of ABO blood group among various populations

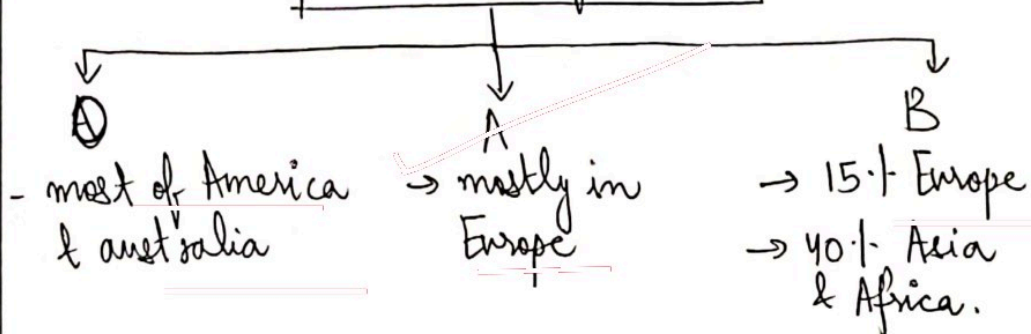


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Phenotype of ABO  $\rightarrow$  A, B, AB, O

Genotype of ABO  $\rightarrow$  AA, AO, AB, BB, BO, OO

### Prevalance of ABO



### Applications of ABO

paternity dispute  
detection of genetic disorder  
facial reconstruction.

### 2) Rh system

- It has eight haplotypes and different alleles at different locuses of chromosomes.
- Rh- is almost rare & found in some parts of Europe Africa.

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- Rh1 is found ~~in~~ majorly in Africa.
- Rh2 is found in South Asia.

Ⓐ Rh blood group is also helpful in ~~determining~~ determining whether a mother have ~~a~~ erythroblastis fetalis.

Hence, genetic markers have immense application in applied genetics and are also helpful in detecting various disease.

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